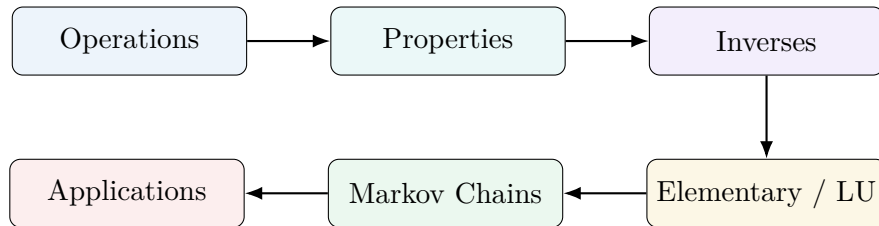


Chapter 2: Matrices

Sections 2.1–2.6

Operations, Inverses, Factorizations, Markov Chains, and Applications



MATH 3260 – Linear Algebra

Original examples with selected Spurs, music, hiking, badminton, driving, and travel themes.

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1 Section 2.1: Operations with Matrices

Learning objectives

Students will identify matrix size and entries, test equality, add and scale matrices, multiply matrices, write matrix equations, and interpret matrix products as column combinations.

1.1 Matrices, entries, and equality

Matrix and size

An $m \times n$ matrix has m rows and n columns. Its entry in row i , column j is a_{ij} . Two matrices are equal exactly when they have the same size and equal corresponding entries.

Equality with a parameter

Let

$$A = \begin{bmatrix} 9 & 20 \\ 21 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 9 & p \\ 21 & 1 \end{bmatrix}.$$

Then $A = B$ exactly when $p = 20$.

1.2 Addition, subtraction, and scalar multiplication

For matrices of the same size,

$$A + B = [a_{ij} + b_{ij}], \quad cA = [ca_{ij}].$$

Fictional Spurs fan-shop inventory

Let

$$K_1 = \begin{bmatrix} 9 & 20 & 21 \\ 12 & 18 & 15 \end{bmatrix}, \quad K_2 = \begin{bmatrix} 11 & 16 & 14 \\ 8 & 21 & 20 \end{bmatrix}.$$

Then

$$K_1 + K_2 = \begin{bmatrix} 20 & 36 & 35 \\ 20 & 39 & 35 \end{bmatrix},$$

and

$$K_1 + 2K_2 = \begin{bmatrix} 31 & 52 & 49 \\ 28 & 60 & 55 \end{bmatrix}.$$

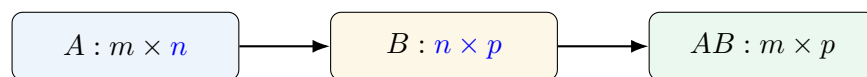
Compatibility

Addition and subtraction require equal sizes. A scalar multiple is always defined, but matrix multiplication has a different compatibility rule.

1.3 Matrix multiplication

If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$, and

$$(AB)_{ij} = \sum_{k=1}^n a_{ik}b_{kj}.$$



The inner dimensions must agree.

Figure 1: Dimension rule for matrix multiplication.

Badminton training plans

Let

$$D = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad S = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 3 \end{bmatrix}.$$

For the first two entries,

$$(DS)_{11} = 1(1) + 2(2) + 0(1) = 5,$$

$$(DS)_{12} = 1(0) + 2(1) + 0(3) = 2.$$

The full product is

$$DS = \begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix}.$$

1.4 Matrix equations and column combinations

Let

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 9 \\ 20 \\ 21 \end{bmatrix}.$$

Then

$$A\mathbf{x} = 9 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + 20 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 21 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 50 \\ 38 \\ 62 \end{bmatrix}.$$

Thus $A\mathbf{x}$ is a linear combination of the columns of A .

1.5 Partitioned matrices

A matrix may be divided into compatible blocks. For example,

$$A = \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{array} \right] = \begin{bmatrix} A_1 & A_2 \end{bmatrix}.$$

Block notation is useful for structured calculations and multiple right-hand sides.

Closing note

Addition combines compatible arrays. Multiplication composes relationships. The matrix equation $A\mathbf{x} = \mathbf{b}$ links matrix algebra, linear systems, and column combinations.

1.6 Definitions and terminology

Term	Meaning
Matrix	Rectangular array of entries.
Size / order	Number of rows by number of columns.
Entry a_{ij}	Entry in row i , column j .
Row / column vector	Matrix with one row / one column.
Scalar multiple	Every entry multiplied by one scalar.
Matrix product	Row-column multiplication when inner dimensions match.
Matrix equation	Equation such as $A\mathbf{x} = \mathbf{b}$.
Partitioned matrix	Matrix divided into blocks.
Linear combination	Weighted sum of vectors or columns.

2 Section 2.2: Properties of Matrix Operations

Learning objectives

Students will use algebraic properties, identity and zero matrices, solve matrix equations, analyze multiplication order, and compute transposes.

2.1 Addition and scalar properties

For compatible matrices and scalars,

$$A + B = B + A, \quad A + (B + C) = (A + B) + C,$$

$$A + O = A, \quad A + (-A) = O,$$

$$c(A + B) = cA + cB, \quad (c + d)A = cA + dA,$$

$$(cd)A = c(dA), \quad 1A = A.$$

Solving a matrix equation

Solve $3X + A = B$ with

$$A = \begin{bmatrix} 1 & 9 \\ 20 & 21 \end{bmatrix}, \quad B = \begin{bmatrix} 10 & 30 \\ 41 & 42 \end{bmatrix}.$$

Then

$$3X = B - A = \begin{bmatrix} 9 & 21 \\ 21 & 21 \end{bmatrix},$$

so

$$X = \frac{1}{3}(B - A) = \begin{bmatrix} 3 & 7 \\ 7 & 7 \end{bmatrix}.$$

2.2 Multiplication properties

When sizes are compatible,

$$A(BC) = (AB)C,$$

$$A(B + C) = AB + AC, \quad (A + B)C = AC + BC,$$

$$c(AB) = (cA)B = A(cB).$$

Associativity, both groupings

Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix}, \quad (AB)C = \begin{bmatrix} 10 & 7 \\ 6 & 7 \end{bmatrix}.$$

Also,

$$BC = \begin{bmatrix} 2 & 1 \\ 4 & 3 \\ 2 & 4 \end{bmatrix}, \quad A(BC) = \begin{bmatrix} 10 & 7 \\ 6 & 7 \end{bmatrix}.$$

Multiplication is not commutative

Let

$$P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

Then

$$PQ = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad QP = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Therefore, $PQ \neq QP$.**2.3 Identity, zero, and transpose**

The identity satisfies

$$AI_n = A, \quad I_m A = A.$$

The transpose interchanges rows and columns. If

$$M = \begin{bmatrix} 9 & 20 & 21 \\ 1 & 5 & 7 \end{bmatrix},$$

then

$$M^T = \begin{bmatrix} 9 & 1 \\ 20 & 5 \\ 21 & 7 \end{bmatrix}.$$

Key identities are

$$\begin{aligned} (A^T)^T &= A, & (A + B)^T &= A^T + B^T, \\ (cA)^T &= cA^T, & (AB)^T &= B^T A^T. \end{aligned}$$

Closing note

Matrix addition behaves much like ordinary addition. Matrix multiplication is associative and distributive, but not generally commutative. Transposing a product reverses the order of the factors.

2.4 Definitions and terminology

Term	Meaning
Zero matrix	Matrix whose entries are all zero.
Additive inverse	$-A$, satisfying $A + (-A) = O$.
Identity matrix	Square matrix with diagonal ones and other entries zero.
Associative	Grouping does not change the result.
Distributive	Multiplication distributes across addition.
Commutative	Order can be reversed; usually false for matrix products.
Transpose	Matrix obtained by switching rows and columns.
Symmetric matrix	Square matrix satisfying $A^T = A$.

3 Section 2.3: The Inverse of a Matrix

Learning objectives

Students will define invertibility, compute inverses by formulas and Gauss-Jordan elimination, use inverse properties, and solve systems with inverses.

Inverse

A square matrix A is invertible if there is a matrix A^{-1} such that

$$AA^{-1} = A^{-1}A = I.$$

If no inverse exists, A is singular. An inverse, when it exists, is unique.

3.1 The 2×2 formula

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

A is invertible if and only if $ad - bc \neq 0$, and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

A determinant-one inverse

Let

$$A = \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}.$$

Since $3(3) - 2(4) = 1$,

$$A^{-1} = \begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix}.$$

Verification:

$$AA^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

3.2 Gauss-Jordan inversion in full detail

Let

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Begin with

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array} \right].$$

Use $R_3 \leftarrow R_3 - R_1$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \end{array} \right].$$

Use $R_3 \leftarrow R_3 + R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 & 1 & 1 \end{array} \right].$$

Use $R_3 \leftarrow \frac{1}{2}R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array} \right].$$

Use $R_2 \leftarrow R_2 - R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array} \right].$$

Use $R_1 \leftarrow R_1 - R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array} \right].$$

Therefore,

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}.$$

Singularity

For

$$S = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix},$$

$ad - bc = 0$, and Row 2 is twice Row 1. The matrix is singular.

3.3 Inverse properties

For invertible matrices,

$$\begin{aligned} (A^{-1})^{-1} &= A, & (cA)^{-1} &= \frac{1}{c}A^{-1}, \\ (A^T)^{-1} &= (A^{-1})^T, & (AB)^{-1} &= B^{-1}A^{-1}. \end{aligned}$$

3.4 Solving a system with an inverse

Solve

$$A\mathbf{x} = \begin{bmatrix} 29 \\ 41 \\ 30 \end{bmatrix}.$$

Using the inverse above,

$$\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 29 \\ 41 \\ 30 \end{bmatrix}.$$

Thus

$$\begin{aligned}x_1 &= \frac{1}{2}(29 - 41 + 30) = 9, \\x_2 &= \frac{1}{2}(29 + 41 - 30) = 20, \\x_3 &= \frac{1}{2}(-29 + 41 + 30) = 21.\end{aligned}$$

Therefore,

$$\mathbf{x} = \begin{bmatrix} 9 \\ 20 \\ 21 \end{bmatrix}.$$

Closing note

The Gauss-Jordan method exposes invertibility directly: the left block becomes I exactly when the inverse exists. The same inverse can solve many systems sharing the same coefficient matrix.

3.5 Definitions and terminology

Term	Meaning
Invertible / nonsingular	Matrix with a multiplicative inverse.
Singular	Matrix without an inverse.
Inverse	Matrix satisfying $AA^{-1} = A^{-1}A = I$.
Adjoining I	Forming $[A \mid I]$ for inversion.
Inverse of a product	$(AB)^{-1} = B^{-1}A^{-1}$.
Cancellation	Valid when the cancelled matrix is invertible.

4 Section 2.4: Elementary Matrices and LU-Factorization

Learning objectives

Students will construct elementary matrices, connect row operations to left multiplication, factor matrices into elementary matrices, compute LU-factorizations, and solve systems by triangular substitution.

Elementary matrix

An elementary matrix is obtained from an identity matrix by one elementary row operation. Left multiplication performs that same row operation on any compatible matrix.

4.1 Three types

From I_3 :

$$E_{\text{swap}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$E_{\text{scale}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$E_{\text{replace}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}.$$

For

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix},$$

$$E_{\text{replace}}A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 3 & 2 \end{bmatrix},$$

which performs $R_3 \leftarrow R_3 - 2R_1$.

4.2 Product of elementary matrices

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - 3R_1$, then $R_1 \leftarrow R_1 - 2R_2$. The elementary matrices are

$$E_1 = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}.$$

Since $E_2E_1A = I$,

$$A = E_1^{-1}E_2^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

4.3 LU-factorization in full detail

Consider

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 4 & 5 & 0 \\ -2 & 8 & 11 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - 2R_1$:

$$\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ -2 & 8 & 11 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 + R_1$:

$$\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 9 & 10 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - 3R_2$:

$$U = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix}.$$

Store multipliers 2, -1, 3 below the diagonal:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}.$$

Therefore, $A = LU$.

4.4 Solving with LU

For

$$\mathbf{b} = \begin{bmatrix} -9 \\ 49 \\ 290 \end{bmatrix},$$

solve $L\mathbf{y} = \mathbf{b}$:

$$y_1 = -9,$$

$$2y_1 + y_2 = 49 \Rightarrow y_2 = 67,$$

$$-y_1 + 3y_2 + y_3 = 290 \Rightarrow y_3 = 80.$$

Then solve $U\mathbf{x} = \mathbf{y}$:

$$4x_3 = 80 \Rightarrow x_3 = 20,$$

$$3x_2 + 2x_3 = 67 \Rightarrow x_2 = 9,$$

$$2x_1 + x_2 - x_3 = -9 \Rightarrow x_1 = 1.$$

Thus

$$\mathbf{x} = \begin{bmatrix} 1 \\ 9 \\ 20 \end{bmatrix}.$$

Closing note

Elementary matrices encode row operations. LU-factorization stores elimination multipliers in L and the resulting echelon matrix in U , enabling efficient repeated solves.

4.5 Definitions and terminology

Term	Meaning
Elementary matrix	Identity matrix after one row operation.
Row equivalent	Related by elementary row operations.
Lower triangular	Entries above the diagonal are zero.
Upper triangular	Entries below the diagonal are zero.
LU-factorization	$A = LU$.
Multiplier	Elimination factor stored in L .
Forward substitution	Solve lower triangular system top-down.
Back substitution	Solve upper triangular system bottom-up.

5 Section 2.5: Markov Chains

Learning objectives

Students will recognize stochastic matrices, compute future states, find steady states, and analyze regular and absorbing chains.

Column-stochastic convention

State vectors are columns. The entry p_{ij} is the probability of moving from state j to state i . Each column sums to 1, and

$$\mathbf{x}_{k+1} = P\mathbf{x}_k, \quad \mathbf{x}_n = P^n \mathbf{x}_0.$$

5.1 Two-state music preference model

Let

$$P = \begin{bmatrix} 0.80 & 0.30 \\ 0.20 & 0.70 \end{bmatrix}, \quad \mathbf{x}_0 = \begin{bmatrix} 0.21 \\ 0.79 \end{bmatrix}.$$

Then

$$\mathbf{x}_1 = P\mathbf{x}_0 = \begin{bmatrix} 0.405 \\ 0.595 \end{bmatrix}.$$

The steady state satisfies $P\mathbf{q} = \mathbf{q}$ and $q_1 + q_2 = 1$. From

$$0.8q_1 + 0.3q_2 = q_1,$$

we obtain $2q_1 = 3q_2$, so

$$\mathbf{q} = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}.$$

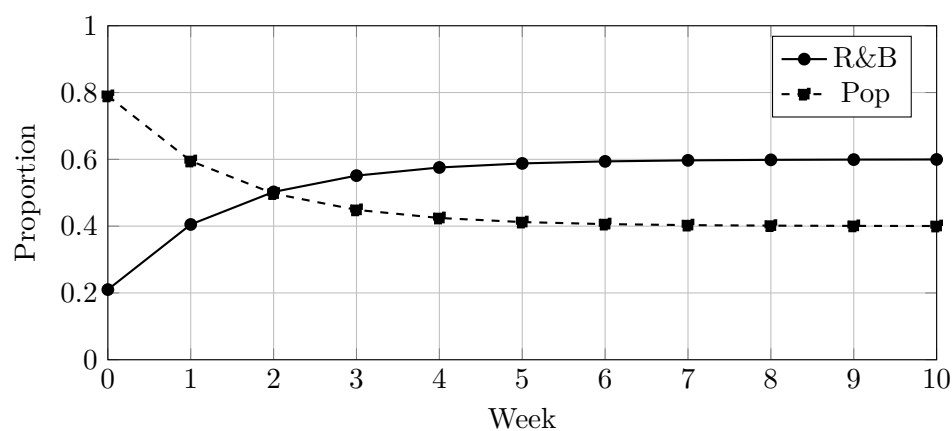


Figure 2: Convergence to the steady state $(0.6, 0.4)^T$.

5.2 Three-state travel-mode model

Let states be driving, transit, and walking:

$$P = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.2 & 0.6 & 0.3 \\ 0.1 & 0.2 & 0.6 \end{bmatrix}, \quad \mathbf{x}_0 = \begin{bmatrix} 0.5 \\ 0.3 \\ 0.2 \end{bmatrix}.$$

The steady state is

$$\mathbf{q} = \begin{bmatrix} 10/29 \\ 11/29 \\ 8/29 \end{bmatrix}.$$

5.3 Absorbing hiking chain

Use states: on trail, summit, returned. Let

$$P = \begin{bmatrix} 0.5 & 0 & 0 \\ 0.3 & 1 & 0 \\ 0.2 & 0 & 1 \end{bmatrix}.$$

The summit and return states are absorbing.

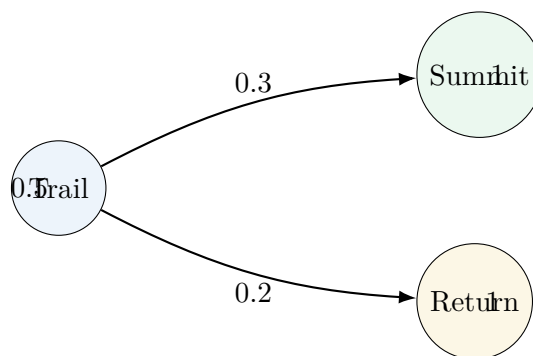


Figure 3: Original absorbing hiking-chain model.

Starting from $\mathbf{x}_0 = (1, 0, 0)^T$,

$$\mathbf{x}_1 = \begin{bmatrix} 0.5 \\ 0.3 \\ 0.2 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 0.25 \\ 0.45 \\ 0.30 \end{bmatrix}.$$

The transient probability is $(1/2)^n$, so

$$\lim_{n \rightarrow \infty} \mathbf{x}_n = \begin{bmatrix} 0 \\ 0.6 \\ 0.4 \end{bmatrix}.$$

Closing note

Matrix powers model repeated transitions. Regular chains converge to a unique steady state, while absorbing chains describe eventual outcomes.

5.4 Definitions and terminology

Term	Meaning
State	Category occupied by a population member.
Transition probability	Probability of moving between states.
Stochastic matrix	Nonnegative matrix whose columns sum to 1.

State vector	Column of probabilities or proportions.
Markov chain	Repeated process with fixed transition probabilities.
Steady state	$\mathbf{q}$ satisfying $P\mathbf{q} = \mathbf{q}$.
Regular matrix	Some power has all positive entries.
Absorbing state	State that cannot be left.
Absorbing chain	Chain that can reach an absorbing state.

6 Section 2.6: More Applications of Matrix Operations

Learning objectives

Students will use matrices in cryptography, Leontief input-output models, and least squares regression.

6.1 Cryptography

Map blank to 0 and A through Z to 1 through 26. Use

$$E = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad E^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}.$$

Encoding GO SPURS GO

The numerical sequence is

$$7, 15, 0, 19, 16, 21, 18, 19, 0, 7, 15, 0.$$

Partition into pairs and multiply on the right by E :

$$[7 \ 15]E = [22 \ 59], \quad [0 \ 19]E = [19 \ 57],$$

$$[16 \ 21]E = [37 \ 95], \quad [18 \ 19]E = [37 \ 93],$$

$$[0 \ 7]E = [7 \ 21], \quad [15 \ 0]E = [15 \ 30].$$

The cryptogram is

$$22, 59, 19, 57, 37, 95, 37, 93, 7, 21, 15, 30.$$

Decoding multiplies each coded row by E^{-1} .

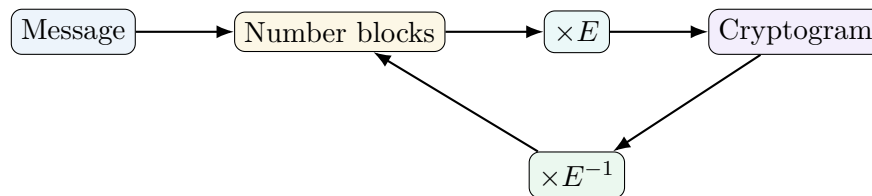


Figure 4: Matrix encoding and decoding workflow.

6.2 Leontief input-output model

For total output $\mathbf{x}$, internal-consumption matrix D , and external demand $\mathbf{e}$,

$$\mathbf{x} = D\mathbf{x} + \mathbf{e},$$

so

$$(I - D)\mathbf{x} = \mathbf{e}.$$

Let

$$D = \begin{bmatrix} 0.10 & 0.05 & 0.10 \\ 0.20 & 0.10 & 0.05 \\ 0.05 & 0.10 & 0.10 \end{bmatrix}, \quad \mathbf{e} = \begin{bmatrix} 95 \\ 62 \\ 56 \end{bmatrix}.$$

Then

$$I - D = \begin{bmatrix} 0.90 & -0.05 & -0.10 \\ -0.20 & 0.90 & -0.05 \\ -0.05 & -0.10 & 0.90 \end{bmatrix}.$$

Multiplying the system by 20 gives

$$\begin{bmatrix} 18 & -1 & -2 \\ -4 & 18 & -1 \\ -1 & -2 & 18 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 1900 \\ 1240 \\ 1120 \end{bmatrix}.$$

The solution is

$$\mathbf{x} = \begin{bmatrix} 120 \\ 100 \\ 80 \end{bmatrix}.$$

Check:

$$D\mathbf{x} + \mathbf{e} = \begin{bmatrix} 25 \\ 38 \\ 24 \end{bmatrix} + \begin{bmatrix} 95 \\ 62 \\ 56 \end{bmatrix} = \mathbf{x}.$$

6.3 Least squares regression

Use the original dataset

$$(1, 8), (2, 9), (3, 17), (4, 17), (5, 24).$$

Fit $y = a_0 + a_1x$ using

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 8 \\ 9 \\ 17 \\ 17 \\ 24 \end{bmatrix}.$$

The normal equations are

$$X^T X \mathbf{a} = X^T \mathbf{y},$$

with

$$X^T X = \begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix}, \quad X^T \mathbf{y} = \begin{bmatrix} 75 \\ 265 \end{bmatrix}.$$

Thus

$$\begin{cases} 5a_0 + 15a_1 = 75, \\ 15a_0 + 55a_1 = 265. \end{cases}$$

Multiply the first equation by 3 and subtract:

$$10a_1 = 40 \Rightarrow a_1 = 4.$$

Then $a_0 = 3$, so

$$\hat{y} = 3 + 4x.$$

The residual vector is

$$\mathbf{r} = \begin{bmatrix} 1 \\ -2 \\ 2 \\ -2 \\ 1 \end{bmatrix},$$

and

$$\text{SSE} = 14.$$

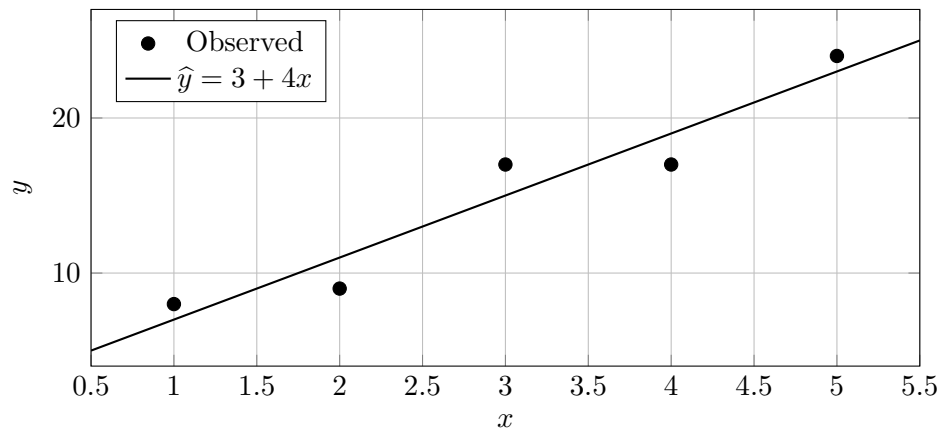


Figure 5: Least squares line and original data.

Closing note

The same matrix operations support reversible encodings, economic balance models, and best-fit approximations. Applications differ, but the algebraic structure is shared.

6.4 Definitions and terminology

Term	Meaning
Cryptogram	Encoded message.
Encoding / decoding matrix	Invertible transform and its inverse.
Input-output matrix	Internal industry consumption coefficients.
External demand	Demand outside the modeled industries.
Least squares line	Line minimizing squared residuals.
Design matrix	Matrix containing regression predictor structure.
Normal equations	$X^T X \mathbf{a} = X^T \mathbf{y}$.
Residual	Observed minus predicted value.
SSE	Sum of squared errors.

7 Chapter 2 Summary

Conceptual chain

operations $\longrightarrow$ properties $\longrightarrow$ inverses
 $\longrightarrow$ elementary matrices and LU
 $\longrightarrow$ Markov and application models.

Always check dimensions, preserve product order, show intermediate matrices, verify computational results, and interpret answers in context.